Contact

Astana
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www.linkedin.com/in/kirilovu (LinkedIn)
aipush.dev/ (Portfolio)

Top Skills

Artificial Intelligence (AI)
Retrieval-Augmented Generation (RAG)
RabbitMQ

Languages

English (Limited Working)
Русский (Native or Bilingual)
Белорусский (Native or Bilingual)

Aliaksandr Kirylau

AI Agent Engineer | Lead Systems & Data Engineer | Java · Python | High-Load · Kafka · Redis · Postgres · ClickHouse | LangGraph · Agentic Swarms · MCP
Mafra, Lisbon, Portugal

Summary

Staff Systems Architect and Data Engineer with 15+ years of experience building extreme-scale platforms for product companies (Odnoklassniki, Wargaming, REDCap Cloud). Take full ownership of complex architectural challenges: from initial system design and PoC to production-ready deployment, DevOps collaboration, observability, and incident resolution. Bridging cutting-edge Agentic AI with the rigorous discipline of high-load distributed engineering.Big Data & High-Load Streaming- Social Network Scale: Built core notification and rate-limiting engines handling 40M DAU and 350K req/sec.- Enterprise ETL & Streaming: Architected real-time CDC pipelines processing 250M+ clinical records.- Stack: Java, Apache Kafka, Kafka Connect, Debezium, Redis, ClickHouse, PostgreSQL.AI & Agentic Systems- PsychoAI Platform: Designed and shipped a production B2T clinical copilot.- Architecture: Hybrid multi-agent swarms with secure MCP data integrations.- LLMOps: RAG over sensitive data, automated LLM-as-a-Judge evaluations, and privacy-first telemetry.- AI-Driven Development (AIDD): 2+ years integrating autonomous agents across the SDLC for advanced code generation, CI/CD reviews, and real-time production metric analysis.- Stack: Python, LangGraph, CrewAI, MCP, Qdrant, Langfuse, DSPy.Leadership & E2E Delivery- Led engineering teams (up to 10 people), partnering directly with business stakeholders.- Cloud-native delivery: Kubernetes (AWS/GCP) rollouts, metric gates, and runtime kill-switches.English - B2 (Professional working proficiency)

Experience

PsychoAI
AI Systems Architect
January 2026 - Present (10 months)
Warsaw, Mazowieckie, Poland

An innovative LLM-powered copilot designed to elevate the quality of psychotherapy.

The core engine transcribes sessions, leverages LLMs to analyze the clinical picture, suggests therapeutic directions, and automatically detects high-risk edge cases (e.g., suicidal ideation).

- Collaborated directly with practicing psychotherapists to design and build a system that effectively solves real clinical workflows.
- Built a scalable multi-agent Python backend. Integrated LangGraph nodes with CrewAI swarms, granting agents secure MCP access to various data sources (PostgreSQL, ClickHouse, Qdrant, Hybrid RAG (vector search + BM25)) to autonomously generate comprehensive client reports and clinical analyses.
- Designed an intelligent matchmaking system using CrewAI to analyze new client intake data and recommend the most suitable psychotherapists and methodologies.
- Ran ~10,000 synthetic tests across 10 distinct patient personas and 4 psychotherapy methodologies to rigorously optimize LLM prompts and ensure clinical safety.

Stack: Python, FastAPI, LangGraph, CrewAI, Langfuse, Redis, PostgreSQL, ClickHouse, Qdrant, MCP, Kubernetes.

REDCap Cloud
Staff Software Engineer / Team Lead
December 2021 - Present (4 years 11 months)
Atlantic Beach, Florida, United States

REDCap Cloud is a certified global data platform for clinical trials, trusted by Pfizer and other leading pharmaceutical companies.

As a Team Lead and Architect, I led the platform's adaptation for Pfizer's scale - building a comprehensive, ClickHouse-powered analytics and reporting system.

- Architected a high-load streaming ETL pipeline (PostgreSQL CDC, Kafka, ClickHouse) to power a complex visual reporting interface.
- Engineered a sophisticated authorization model directly within ClickHouse to enforce strict data visibility scopes, role-based access, and dynamic data masking.

- Restructured native backend services for Kubernetes cloud deployment, enabling automatic scaling and resilient execution.
- AI-Driven Development (custom MCP skills/rules, architecture generation, and token optimization).
- Production monitoring, incident investigation, local k8s environments, developer onboarding.

Stack: Java (Springboot, Hibernate, Quartz), Kafka/Kafka Connect, Debezium, ClickHouse, PostgreSQL, RabbitMQ, Kubernetes.

Wargaming
Staff Software Engineer / Team Lead
June 2014 - December 2021 (7 years 7 months)
Lithuania

Backend engineering for Big Data and ML integration platforms across Wargaming's AI-powered products.

As a Technical Lead, I led backend teams and designed the core integration architectures.

- Designed the integration architecture connecting Python ML modules and AWS SageMaker into Java-based high-load backend services.
- Built data pipelines ingesting social platform APIs (Instagram, YouTube, TikTok, Facebook) at massive scale.
- Implemented sentiment analysis pipelines for influencer audience comments and demographic profiling.
- Engineered a company-wide dynamic configuration framework allowing runtime feature toggles without system redeploys.
- Proposed and delivered a self-serve, token-based monetization system that opened a major new revenue channel for the product.

Stack: Java (Spring Boot), Python, AWS (SageMaker, SQS, S3), PostgreSQL, MongoDB, Redis, RabbitMQ.

Odnoklassniki. Mail.Ru Group.
Senior Software Engineer
May 2012 - July 2014 (2 years 3 months)
Riga, Latvia

Core platform engineering for one of the largest social networks in Europe (40M+ DAU, 350,000 req/sec peak load).

- Engineered the core Game API platform, serving as the high-load bridge between 140M active users and 10,000 third-party applications.
- Architected a smart notification engine with custom rate-limiting, transforming legacy mass-mailing into targeted delivery (Spam ↓ 80%, CTR ↑ 6%).
- Developed a predictive game recommendation engine based on user gameplay history, establishing the foundation for a personalized smart game catalog.
- Won multiple internal hackathons by prototyping pioneering features (Live Avatars, Secret Chats, HTML5 Notifications) in 2012, anticipating their mainstream adoption in modern messengers.

Stack: Java, MS SQL Server, Cassandra, Tomcat.

CrowdComputing Systems, Inc.
Senior Java Developer, TL
August 2011 - April 2012 (9 months)
Минск

Pioneering smart crowdsourcing platform for automated, large-scale ML data labeling via Amazon Mechanical Turk.

- Engineered a data pipeline that ingested massive photo datasets and orchestrated ~1M micro-labeling tasks on MTurk.
- Implemented automated cross-worker validation to ensure high-fidelity data quality.
- Delivered the clean, foundational labeled datasets required for training large-scale ML and early AI/LLM models.
- Powered search and image recognition engines for top-tier enterprise clients, including Microsoft Bing.

Stack: Java (Springboot), MySQL, Amazon Mechanical Turk API.

Exadel
Senior Java Developer, Team Lead
October 2010 - July 2011 (10 months)
Minsk

Developed modules for Next Wave Logistics - a large-scale enterprise E-CRM system for direct sales businesses. Led a team for 2 months, handled

architecture design, customer communication, and delivery of commission processing and awards management modules.

Stack: Java · Spring MVC · Hibernate · MS SQL · ExtJS

SoftTeco
Java Developer
May 2008 - May 2009 (1 year 1 month)

Developed Ray - a payment proxy system for online merchants. The system selected optimal payment channels based on card type and transaction parameters, routing funds directly to merchants while reducing standard transaction fees.

Stack: Java EE · EJB3 · Hibernate · JSF · PostgreSQL · Glassfish

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Education

Belaruski Dzjaržauny Universitet Informatyki i Radyjoelektroniki
Bachelor of science, System Engineer · (2001 - 2006)